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| **Fiscal Year:** | FY 2006 | **Task Last Updated:** | FY 12/23/2005 |
| **PI Name:** | Zimmerberg, Joshua M.D., Ph.D. | | |
| **Project Title:** | Bioreactor Technology Transfer | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | BIOTECHNOLOGY--Biotechnology | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 20892-1855 | **Congressional District:** | 8 |
| **Comments:** | | | |
| **Project Type:** | Flight,Ground | **Solicitation / Funding Source:** | 2001 Physical Sciences 01-OBPR-08 |
| **Start Date:** | 02/26/1999 | **End Date:** | 09/30/2007 |
| **No. of Post Docs:** | 7 | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | 24 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | 3 | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | 6 | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | ISS | | |
| **Flight Assignment:** | ISS<br>NOTE: Funding reinstated to 9/30/2007, per J. Cohen (HQ, 8/2006) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | H30993D | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

**Task Description:**

The NASA-NIH Center for Three-Dimensional Tissue Culture supports a number of scientists who work on projects related mainly to cell-cell interactions, cell differentiation, and the pathogenesis of infectious diseases. Many mainstream medical research studies have important implications for human space exploration and the Center helps promote the transfer of technology and interests between NASA and NIH. Intramural and extramural scientists launch pilot projects involving the RWV Bioreactor as extensions of their main research projects. Based on preliminary results, a research group can apply for a small grant proposal for a study involving the RWV. Additionally, many in-house projects are ongoing at the Center.
Below are the current projects at the NASA/NIH Center:

Study of immune dysfunction of human lymphoid tissue in microgravity

Study of viral reactivation in ex vivo human lymphoid tissue

Models of mucosal transmission of HIV

Effect of microgravity on cellular membrane reorganization

Effect of culture conditions, especially suspension, on cell membrane properties

A large-scale method of malaria parasite production in Rotating Hollow Fiber Cell Bioreactor

Study of the effect of magnetic field on malaria parasite replication in human red blood cells

Extracellular signals on differentiation of embryonic stem cells

Mesenchymal stem cells in wound healing and the role of apoptotic pathways

Effect of microgravity on phosphatidylserine exposure and myoblast-myoblast fusion

The goal of the Center is to maintain a core facility at NIH for use of the NASA Rotating Wall Vessel (RWV) Bioreactor and related three-dimensional tissue culture systems for studying the basic mechanisms of normal and pathological intercellular interactions in multi-cellular systems, and how these interactions can be influenced by modeled microgravity. Investigators provide biological material or instruct the Center staff on experiment design, dissection and culture techniques. The investigator provides mentorship and is available for questions beyond staff expertise. The Center researchers prepare and culture the cells or tissues, and perform the primary analysis, including gathering data on tissue metabolism and tissue cytoarchitecture. The Center's staff also provides services such as immunocytochemistry, ELISA, confocal microscopy, electron microscopy, and flow cytometry. The primary investigator performs specialized analyses specific to their project. Successful pilot projects can then apply for funding, which keeps the staff of the Center available for new pilot projects, and promotes technology transfer by recruiting scientists to work on three-dimensional tissue culture full time. Because of our NASA funding, and because of the possibility that results obtained in suspension culture may relate to health problems of astronauts, or present unique opportunities for studying physiology in microgravity, there are possibilities of flight experiments for PIs of NASA/NIH projects. Currently we have flown one cellular experiment on the International Space Station (ISS), and we are currently running fluid dynamics experiments on the ISS.

The Center provides researchers an opportunity to develop new model systems for biomedical research and provides access to the RWV, which allows cellular growth, aggregation, and development of in vivo-like structures. This vessel allows the culture of cells under minimal shear forces in a well-oxygenated medium under conditions that mimic microgravity. RWVs and experienced staff are available for testing tissues, primary cell cultures, and cell lines under these conditions that facilitate cell-cell interactions and promote differentiation. NASA funding for equipment has enabled us to procure state-of-the-art microscopes to determine structure and function relationships between cells of developed tissues. The design of the Center allows investigators to assess potential benefits of the RWV for their specific aims without having to commit major resources, which greatly increases the number of investigators who will explore the RWV as a tissue culture opportunity and apply their knowledge and experience to space research. Results from projects at the Center benefit space research, since the RWV models some aspects of microgravity, but since they stem from basic medical research, they have many terrestrial benefits as well.

: The impact of the work conducted at the NASA/NIH Center has broad impact on all of these levels. First, we are closer to a cure for AIDS and HIV disease as a result of our work. We finally understand more about the pathogenesis of the morbidity and mortality caused by this infection. Second, we are closer to curing Lyme disease because we finally have a model system to test for gene expression. Third, we understand better now how to use embryonic stem cells. Fourth, we know how the flu works to create patches on membranes for the virus to bud from, and how the invasion of an organism by the flu is cholesterol-dependent. Fifth, we begin to understand how measles inhibits HIV. Sixth, we start to understand better the entry of HIV through the cervix. Seventh, we have discovered how Insulin works to stimulate glucose transport. This will lead to better therapeutics for Diabetics. Eighth, we have discovered how the malaria parasite is released at the end of one cycle to begin another. This will help all of the infected people in the world, one of which dies every 10 seconds. Because of the enormous economic impact of flu, HIV, malaria, and diabetes, this work has the potential to allieviate human suffering and have a huge impact on the economies of many nations, including the United States of America, in its new campaign against the flu pandemic.

The impact of our FDI experiment on the ISS is also of enormous importance. It is the first time we have been able to carry out research on the station that is interactive and iterative with the scientist-astronaut playing the crucial intellectual role. It is the first time we have been able to see what the specimens look like in great detail, and we discovered a plague of bubbles in all of our specimen chambers. The solution of this problem is needed in order to understand the behavior of almost all fluids on the ISS, and in future space flight. In addition, once we get this data we will be able to improve greatly our ability to perform scientific experiments in space. This will let us find out once and for all if microgravity has adverse effects on astronauts lives.

**Rationale for HRP Directed Research:**

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| **Research Impact/Earth Benefits:** | Since most of the studies conducted at the Center stem from mainstream medical research, the results obtained benefit not only space research, but the entire medical research community. In particular, the pathogenesis studies (HIV, malaria, viral reactivation, influenza) and stem cell studies (embryonic, mesenchymal) all have important Earth benefits. Studies on cell membrane properties, cell fusion, and apoptosis benefit the entire scientific community by providing insight to basic cell functions. |
| **Task Progress:** | To support human exploration of space, it is essential that we assure that the highest standards of medical care be available to astronauts, even during space flights of extended duration. Towards that aim, it has been invaluable to the mission of space exploration to have a Center that promotes the exchange of technology and interests between NASA and NIH. In 2005, intra- and extra-mural scientists worked at the NASA/NIH Center for Tissue Culture on ten various projects related to the effects of microgravity and modeled microgravity (in the NASA Rotating Wall Vessel Bioreactor (RWV)) on both physiology and pathophysiology of human cells and tissues, as well as developing the means for studying the behavior of various types of stem cells.<br>During 2005, many advances have been made in the ongoing projects, and several new projects have been initiated. The Center scientists have continued to focus on flight experiments on Increments 7, 8, and 10 onboard International Space Station (ISS) to optimize fluid/particle mixing in the cell culture chambers, as well as develop bubble removal procedures, both of which will improve culture conditions allowing optimal science to be conducted.<br><br>Toward the goal of understanding immune dysfunction of human lymphoid tissue in microgravity, new studies are being designed to look at B cell dysfunction, and to focus on supra-molecular activation complexes in lymphocyte activation in modeled microgravity. Additionally we are making great progress on an alternate model of immune impairment (which demonstrates similar responses to those in modeled microgravity) and have determined that there is a soluble factor produced by lymphoid tissue that suppresses B cell function.<br><br>New studies on viral reactivation show that human lymphoid tissue is a good model for studying the reactivation of certain latent viral infections and using this tissue in RWVs will, to some extent, mimic the effects of microgravity on viral reactivation, by the induction of immune suppression or by mechanisms yet to be discovered. For the study on models of mucosal transmission of HIV-1, a comprehensive system for pre-clinical evaluation of potential microbicides against HIV-1 is using information accrued from several different model systems (lymphoid tissue, cervix, and rectal tissue). Another aspect of the system is to address host factors that influence the outcome of disease in immunocompromised tissues, which has implications not only for HIV disease but also for space-related diseases, such as the secondary immunodeficiency disease seen in astronauts.<br><br>Studies on the effect of microgravity on cell membrane organization are investigating the relationship between membrane lipids and proteins, and raft organization on the surface of influenza virus; changes in the raft organization of the cell membrane may be responsible for immune impairment. Related studies on cell properties in physical suspension focus on determining whether hemagluttinin (HA) surface expression in HAb2 cells, and the Hab2 cells' fusogenic properties are dependent on cell culture conditions, and evaluating any changes in cell contact inhibitions. Changes in these characteristics have potentially serious impact on issues of astronaut health, such as increased vulnerability to infection and delayed wound healing. We have developed a new method of malaria parasite culture in a hollow fiber cell bioreactor that significantly increased the yield of infected cells and reduced labor. This could lead to a prototype method for culturing highly metabolically active organisms in a closed system with minimal intervention by the researchers suitable for space flights applications.<br><br>An important and not well-studied factor of space flight is the exposure of humans to various space and equipment generated magnetic fields (MF), so new studies were initiated to study the effect of MF on a unique system, the malaria parasite, that has magneto sensitive (paramagnetic) structures inside red blood cells (RBC). Embryonic stem cell (ESC) studies are showing that the composition and topology of the extracellular matrix (ECM) determine the pattern of ESC differentiation by changing cell shape or/and growth factor responsiveness. ESCs in the context of 3D ECMs of various compositions and topology exhibit different cell morphology, patterns of differentiation, and different degrees of sensitivity towards growth factors. Another new study is on human mesenchymal stem cells (hMSC) in wound healing and the role of apoptotic pathways. Our results show differential apoptotic responses in hMSCs in modeled microgravity (RWV bioreactor) versus normal conditions and we are testing the hypothesis that stem cells counterattack activated lymphocytes by using the FasL-mediated apoptotic pathway.<br><br>Another new study is on the effect of microgravity on phosphatidylserine exposure and myoblast-myoblast fusion; we are testing whether fusion of two distinct cell membranes depend on Phosphatidylserine exposure and to understand how modeled microgravity affects myoblast-myoblast fusion. |

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